Ionic & covalent bonds

Form ions

1 Electron transfer

Metal atoms lose electrons → positive ions. Non-metals gain → negative ions. Protons stay unchanged.

Group 1/2: +1/+2 ions; Group 7/6: −1/−2 ions. Electron arrangements match a noble gas.

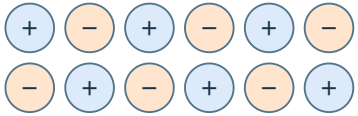
2 Charge → formula

Balance total charges: $Mg^{2+} + 2Cl^{-} \rightarrow MgCl_2$; $2Na^{+} + O^{2-} \rightarrow Na_2O$.

Dot-and-cross ionic diagrams: brackets, ion charge and full outer shells; distinguish electron origins.

Build lattices

3 Ionic lattice



Opposite ions; 2D slice of lattice

A giant, repeating 3D arrangement of opposite ions, with strong electrostatic attractions in all directions.

Many attractions need much energy to overcome → high melting/boiling. Not separate molecules.

4 Conductivity follows movement

Solid: ions held in fixed positions → cannot carry charge through it.

Molten/dissolved: ions free to move → conducts. Electrons do not carry current through the ionic liquid.

Share electrons

5 Covalent pairs

A single bond shares one pair; double shares two; triple shares three.

H_2, Cl_2, HCl: one single bond. O_2: double. N_2: triple. H_2O: two single; NH_3: three; CH_4: four.

Show lone pairs too: O has two in H_2O; N has one in NH_3; Cl has three.

6 Molecules & models

Strong bonds inside a molecule; weaker intermolecular forces between molecules. Boiling overcomes those forces.

Bigger molecules generally have stronger intermolecular forces. No mobile ions/electrons → no conduction.

2D formulae omit 3D shape; ball-and-stick bonds are not literal rods.